
🚀 AWS CodePipeline – Overview

✅ What is AWS CodePipeline?

AWS CodePipeline is a **fully managed continuous integration and continuous delivery (CI/CD) service** that automates the build, test, and deploy phases of your release process every time there is a code change. This helps you rapidly and reliably deliver features and updates.

🛠️ Key Features

Feature	Description
Automated Workflow	Automates build, test, and deploy steps in your pipeline.
Customizable Stages	Define multiple stages such as Source, Build, Test, Deploy, Approval, etc.
Integrations	Works natively with AWS services (CodeCommit, CodeBuild, CodeDeploy, CloudFormation) and third-party tools (GitHub, Jenkins).
Parallel Actions	Run actions concurrently to speed up pipelines.
Manual Approvals	Add manual approval steps to gate deployments.
Event-driven	Pipelines automatically start when source code changes or on schedule.

🔄 How CodePipeline Works

1. **Source Stage**
   Detects code changes from repositories like CodeCommit, GitHub, or S3.
2. **Build Stage**
   Builds your code using services like AWS CodeBuild or Jenkins.

3. **Test Stage** (Optional)
   Runs automated tests to validate your changes.

4. **Deploy Stage**
   Deploys your application using AWS CodeDeploy, CloudFormation, ECS, or other deployment services.

5. **Approval Stage** (Optional)
   Requires manual approval before continuing to next stage.

📊 Pipeline Components

Component	Description
Pipeline	The workflow that defines stages and actions.
Stage	Logical grouping of actions (e.g., Build or Deploy).
Action	A task performed in a stage (e.g., running a build).
Artifact	Files (code, binaries) passed between stages.

🚀 Typical Use Cases

- Continuous delivery of web applications
- Automated deployment of infrastructure as code
- Multi-account or multi-region deployment pipelines
- Compliance workflows with manual approvals
- Blue/green or canary deployments

🔧 Setting Up a Basic Pipeline

1. **Source**: Connect to your GitHub or AWS CodeCommit repo.

2. **Build**: Use CodeBuild to compile and test your app.
3. **Deploy**: Deploy to AWS Elastic Beanstalk, EC2, ECS, or Lambda.
4. **Approval**: Optional manual approval before production deploy.

🔐 Security

- Uses IAM roles to securely access resources.
- Integrates with AWS KMS for encryption of artifacts.
- Supports cross-account pipelines with IAM trust relationships.

💸 Pricing

- Charged based on the number of active pipelines per month.
- No charge for using integrated services like CodeBuild or CodeDeploy.

🧩 Example Pipeline Workflow

[Source (GitHub)] --> [Build (CodeBuild)] --> [Test (CodeBuild)] --> [Deploy (CodeDeploy)] --> [Approval] --> [Production Deploy]

📈 Benefits

- Fully managed, no infrastructure to maintain
- Easy integration with AWS and third-party tools
- Enables rapid, repeatable deployments

- Supports complex workflows with approvals and parallelism
